Laser Cut Assembly Design

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Introduction

This project is a laser-cut based assembly model created using Autodesk Fusion 360. The design contains multiple individual parts assembled together to form a complete mechanical or aerodynamic structure. The project demonstrates CAD modeling, component-based assembly design, and preparation for laser cutting fabrication.

Software Used

- Autodesk Fusion 360
- Laser Cutting Preparation Tools

Project Objectives

- Design a multi-component assembly using CAD software.
- Prepare individual parts for laser cutting.
- Create accurate dimensions for assembly.
- Develop a lightweight and modular structure.

Components Included in the Design

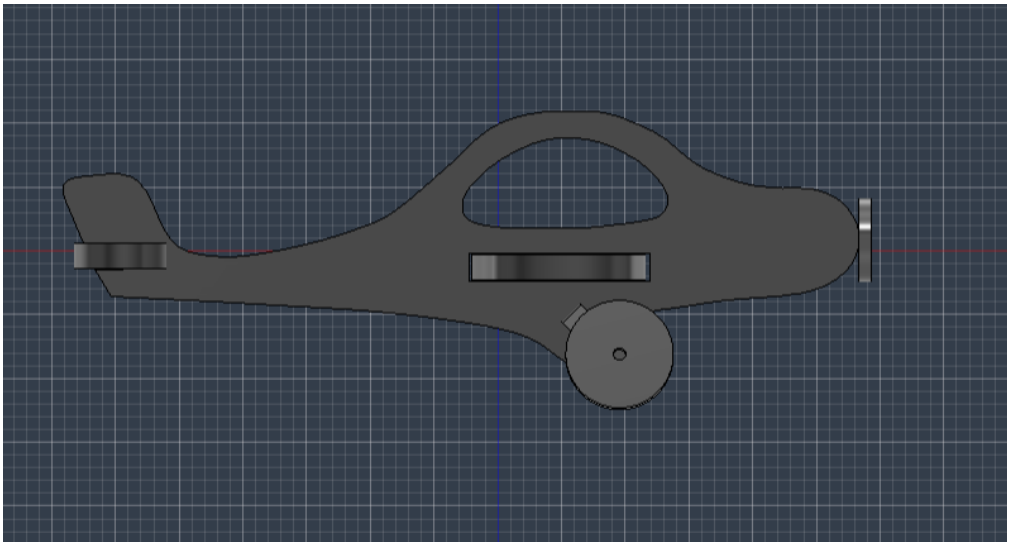
The uploaded Fusion 360 archive contains the following components:

1. Wheel 1
2. Support
3. Body
4. Propeller
5. Wings
6. Tail

These components are assembled together to form the final design.

Project Images

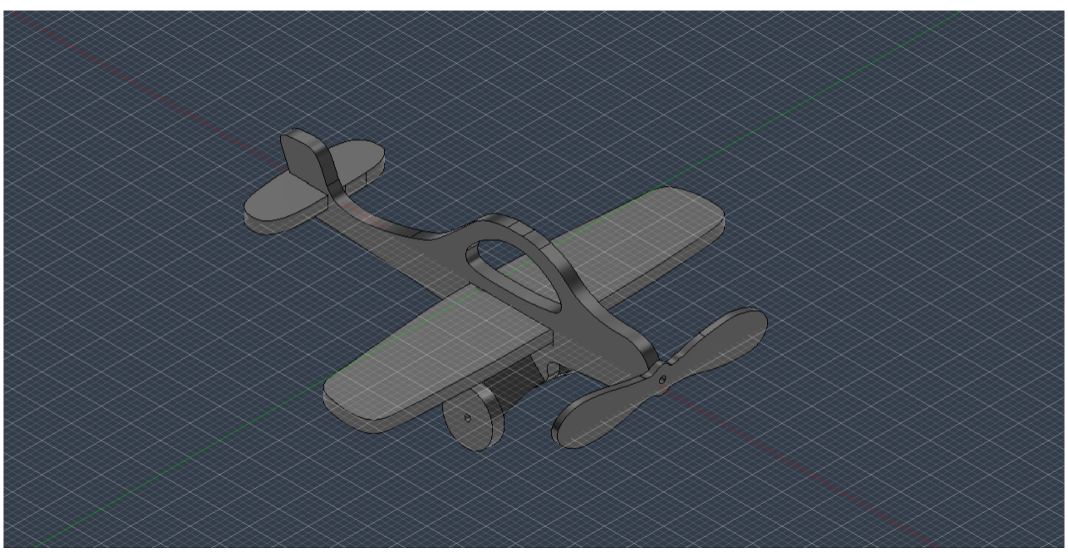
Side View Design



Side View Design

Figure 1: Side view of the laser cut aircraft assembly model showing the body, wheel, tail, and propeller alignment.

Isometric Assembly View



Isometric Assembly View

Figure 2: Complete 3D isometric view of the assembled laser cut aircraft model created in Fusion 360.

Design Description

1. Body

The body acts as the main structural component of the model. It supports and connects all other parts.

2. Wings

The wings are designed to provide balance and aerodynamic appearance to the structure.

3. Tail

The tail section helps stabilize the rear portion of the model.

4. Propeller

The propeller is included as a rotating front component representing propulsion.

5. Wheel

The wheel component provides support for movement or landing functionality.

6. Support

The support structure strengthens the overall assembly and maintains alignment.

Working Principle

The project follows a component-based assembly approach where individual laser-cut parts are manufactured separately and later assembled together using slots, joints, or fasteners.

The design emphasizes:

- Precision fitting
- Lightweight structure
- Easy assembly
- Modular component replacement

Manufacturing Process

1. Create 2D sketches in Fusion 360.
2. Convert sketches into 3D components.

3. Prepare profiles for laser cutting.
4. Export cutting files.
5. Cut components using a laser cutting machine.
6. Assemble all parts together.

Advantages

- High precision fabrication
- Fast manufacturing process
- Lightweight structure
- Easy modification and redesign
- Cost-effective prototyping

Applications

- Educational projects
- Mechanical model prototyping
- Aerodynamic model demonstrations
- STEM learning activities
- CAD and CAM training

Safety Precautions

- Wear protective glasses during laser cutting.
- Ensure proper ventilation.
- Keep flammable materials away from the laser cutter.
- Verify dimensions before cutting.

Conclusion

This laser cut assembly project demonstrates the practical use of Fusion 360 for CAD modeling and laser fabrication. The modular design approach allows accurate manufacturing and easy assembly of components. The project is suitable for educational demonstrations and rapid prototyping applications.

Future Improvements

- Add motorized movement.

- Improve aerodynamic efficiency.
- Use stronger lightweight materials.
- Add electronic control systems.
- Optimize assembly joints for durability.

File Information

Uploaded File Type: Autodesk Fusion 360 Archive (.f3z) Main Assembly File: laser cut.f3d

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Prepared By

Project Documentation Generated from Uploaded Fusion 360 Design File.